Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/18/2006
PI Name:	Conley, Catherine A Ph.D.		
Project Title:	A New Model for Understanding Spaceflight Induced Muscle Atrophy		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Organization Name:	NASA - AMES Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:	NOTE (June 2016): Dr. Conley is currently at NASA Headquarters		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
Start Date:	07/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NCE to 9/30/2007, per A. Chu 6/23/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jacobson, Lewis A (University of Pittsburgh) Szewczyk, Nathaniel J (University of Pittsburgh)		
Grant/Contract No.:	ARC Intramural PI; NNA04CK22A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this project is to develop the nematode C. elegans as a small animal model for studying the molecular basis of spaceflight induced muscle atrophy. C. elegans is an established model both for studying the molecular basis of muscle atrophy on Earth and for studying spaceflight effects on physiology. The principal challenge addressed during this project period is the development, characterization and evaluation of a chemically defined liquid culture system that allows automation of C. elegans experimentation and analysis in flight. We have broken this work into three Aims for the current project period: 1) Adapt the C. elegans muscle protein degradation model to liquid culturing; 2) Validate liquid culturing of C. elegans in the spaceflight environment; 3) Determine biological responses of C. elegans in liquid culture to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1) We have developed a system for the automated culture of the common laboratory animal C. elegans. The system can be used as an enviornmental biomonitor, in pharmaceutical discovery, and in compound toxicity testing. 2) Based upon our work with worm muscles, which lack satellite cells, we believe that altered muscle development may be a contributing factor to human muscle pathologies (in the form of altered development of satellite cells which fuse with developed muscle). If true, this finding opens new avenues of investigation toward muscle pathology treatments. 3) We have identified new genetic triggers of and signaling pathways which regulate protein degradation in worm muscle. These pathways may represent targets for new treatments of human muscle pathologies.
Task Progress:	We have successfully flown C. elegans in our liquid culture system as part of the 2004 Dutch Soyuz Mission. This mission successfully completed the major operational aim (Aim 2) of our project and validated the experimental system; this result has been submitted for publication and is currently under review. Progress on Aim 1 has been slower as we have discovered that C. elegans muscle is different when developed in the chemically defined liquid medium (CeMM). We have established that mutant, transgenic strains of C. elegans grow in CeMM and have determined that gene expression and life history are altered in response to altered diet; these results have been submitted for publication and are currently under review. We have established that the same factors trigger and block muscle protein degradation in CeMM and have found that muscle protein degradation is triggered in response to altered diet (i.e., acute shift from standard diet to CeMM). We have established that while C. elegans are able to adapt life history when developed in CeMM, animals appear unable to adapt when acutely shifted to CeMM and instead catabolize muscle protein as the result; these results are currently being written up for publication. Progress on Aim 3 has been steady and more productive than initially planned. We have confirmed that there are no gross developmental defects in response to spaceflight; this result has been submitted for publication and is currently under review. In collaboration with Dr. Atsushi Higashitani (Tohoku University Graduate School of Life Sciences, Japan) we have shown that checkpoint and physiological apoptosis in germ cells proceed normally in response to spaceflight. This result in particular has relevance to the biological responses to radiation during spaceflight and was published in Apoptosis. In collaboration with Dr. Noriaki Ishioka (Japan Aerospace Exploration Agency) we have demonstrated that muscle development is altered during spaceflight and identified both the functional consequences of this altered development (impaired movement upon reloading) and some of the molecular reasons for this altered development (decreased expression of myogenic transcription factors). This result in particular has relevance to muscle loss experienced in response to spaceflight. As identical molecular changes have been detected in spaceflown C. elegans, chicken cells, rodents, and humans, this result suggests that C. elegans can be employed as a small animal model for studying the effects of spaceflight on muscle. This result is currently in press at the Journal of Experimental Biology. In addition to examining altered gene expression associated with apoptosis and muscle development, we have completed an analysis of the effects of spaceflight on global gene expression. These results are currently being written up for publication.
Bibliography Type:	Description: (Last Updated: 04/15/2008)
Abstracts for Journals and Proceedings	Zhao Y, Youds J, Tarailo M, Tarailo S, Rose A, Selch F, Szewczyk N, Conley CA. "Mutational and genomic response of the nematode Caenorhabditis elgans to a 10 day spaceflight." Platform Talk: Mutagenic Consequences of The Space Environment, COSPAR Colloquium, July 2006. COSPAR Colloquium Proceedings, July 2006. , Jul-2006
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Abstracts for Journals and Proceedings	Baillie D, Rose A, Zhao Y, Higashitani A, Honda S, Honda Y, Kagawa H, Higashibata A, Fujimoto N, Kuriyama K, Ishioka N, Fukui K,Tillman J, Conley CA, Granger L, Segalat L, Gasset G, Eche B, Szewczyk NJ, Chaput D, Viso M. "ICE-FIRST International Ceanorhabditis elegans Experiment first flight on the Space Station." Platform Talk: Science of the European Soyuz Missions to the International Space Station (2001-2005) Congress, June 2006. Science of the European Soyuz Missions to the International Space Station (2001-2005) Congress Proceedings, 2006. , Jun-2006
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Abstracts for Journals and Proceedings	Ishioka N, Higashibata A, Higashitani N, Seki M, Sato M, Szewczyk N, Viso M, Gassett G, Eche B, Fujimoto N, Kuriyama K, Fukui K, Higashitani A. "Microgravity-dependent alteration of protein and gene expression in Caenorhabditis elegans on the international C. elegans space experiment-first (ICE-first)." Poster: 15th International C. elegans Meeting, June 2005. 15th International C. elegans Meeting, Meeting Proceedings, 2005. , Jun-2005
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